Emergency Department Wait Time Prediction Using Machine Learning

Vandan Patel
Department of Computer Science
Georgia Southern University
vp04406@georgiasouthern.edu

Annagrace Howell
Department of Computer Science
Georgia Southern University
ag29516@georgiasouthern.edu

Yasmin Rocio Orduz Landazabal
Department of Computer Science
Georgia Southern University
yo00553@georgiasouthern.edu

***Abstract*—Emergency department (ED) overcrowding is a global healthcare crisis associated with increased patient mortality, reduced care quality, and significant operational inefficiencies. Accurate prediction of patient wait times at the point of triage can empower clinical staff to proactively allocate resources and improve patient communication. This paper presents our systematic literature review (SLR) identifying and synthesizing twelve peer-reviewed research articles (2022–2025) on machine learning approaches for ED wait time and length-of-stay prediction, followed by our data wrangling, exploratory data analysis (EDA), and machine learning pipeline on the Hospital Emergency Room Visits dataset (Kaggle, CC0 license), structured to mirror the MIMIC-IV-ED benchmark. Our dataset comprises 9,216 ED visits with 17 raw features expanded to 26 engineered features after preprocessing. The target variable `ed_los_minutes` has a mean of 189.8 minutes, a median of 153.7 minutes, and a standard deviation of 141.7 minutes. ESI acuity level is the strongest predictor ($r = -0.38$), with mean LOS ranging from 315.7 minutes (ESI 1) to 91.4 minutes (ESI 5). Our review reveals that gradient boosting ensemble methods (XGBoost, LightGBM, Random Forest) consistently achieve the strongest predictive performance. Key gaps identified include the near-universal use of binary classification rather than continuous wait time regression, limited triage-level segmentation, incomplete SHAP-based interpretability, and insufficient fairness analysis. Our project directly addresses these gaps through regression-based ML with SHAP explainability, triage-stratified analysis, and demographic fairness assessment.**

***Index Terms*—emergency department, wait time prediction, machine learning, triage, XGBoost, LightGBM, SHAP, length of stay, patient flow, algorithmic fairness, overcrowding, EDA**

I. Introduction

A. Background and Motivation

Emergency departments serve as the primary point of entry for acute, unscheduled healthcare across the globe. Rising patient volumes, resource constraints, and the unpredictable nature of emergency presentations have made ED overcrowding a recognized public health crisis. Prolonged wait times—defined as the interval between patient arrival and physician evaluation—are directly associated with increased in-hospital mortality, patient dissatisfaction, and a higher rate of patients leaving without being seen (LWBS).

Despite the severity of this problem, most EDs continue to rely on manual triage scoring systems such as the Emergency Severity Index (ESI) and the Manchester Triage Scale (MTS) to prioritize care. While these systems effectively rank urgency, they do not provide quantitative estimates of how long a patient will wait. This gap between urgency classification and wait time transparency is precisely where machine learning (ML) can make a meaningful clinical contribution.

B. Problem Definition

This work is guided by the following central research question: *How can supervised machine learning algorithms trained on routinely collected ED triage data accurately predict patient wait times, and what methodological approaches produce the most clinically interpretable and generalizable models?*

C. Objectives and Novel Contributions

The specific objectives of this project are:

1) Identify and synthesize the most relevant ML research on ED wait time and length-of-stay prediction published between 2022 and 2025.
2) Evaluate the methodological quality, datasets, and model types used across twelve studies.
3) Identify recurring research gaps regarding interpretability, triage-level segmentation, continuous regression, and fairness.
4) Apply regression-based gradient boosting and random forest models to the ED dataset, incorporating SHAP-based feature importance [14], and segment predictions by ESI triage level.

D. Scope

This work covers twelve peer-reviewed articles from reputable sources, all published between 2022 and 2025, and applies the derived insights to a machine learning pipeline. Ten of the twelve articles were published in 2024 or 2025, reflecting the rapidly accelerating pace of research in this domain.

II. Literature Review Methodology

A. Search Strategy

The literature search was conducted systematically across: IEEE Xplore, ACM Digital Library, PubMed/MEDLINE, Springer Link, ScienceDirect (Elsevier), PLOS Digital Health, JAMA Network Open, and Nature/Scientific Data.

B. Keywords Used

The following keyword combinations were used:

- "Emergency department wait time prediction machine learning"
- "ED length of stay ML systematic review"
- "MIMIC-IV-ED machine learning prediction"
- "Triage machine learning emergency department"
- "XGBoost random forest hospital wait time"
- "Interpretable ML healthcare SHAP emergency"
- "ED overcrowding prediction deep learning"
- "Algorithmic fairness emergency department ML"

C. Inclusion and Exclusion Criteria

Table I summarizes the inclusion and exclusion criteria.

TABLE I
INCLUSION AND EXCLUSION CRITERIA

Inclusion Criteria	**Exclusion Criteria**
Published 2022–2025	Published before 2022
English full-text available	Abstracts without full text
Peer-reviewed / top-tier venue	Non-peer-reviewed sources
Supervised ML on ED prediction	ICU/inpatient-only studies
Quantitative metrics reported	No model evaluation
ED wait time, LOS, or triage	Pure administrative studies

III. LITERATURE REVIEW

A. Thematic Overview

The twelve reviewed articles cluster around five major themes: (1) systematic reviews of ML for ED triage and patient flow, (2) empirical ML models for ED wait time and length-of-stay prediction, (3) benchmark datasets and interpretability frameworks, (4) algorithmic fairness and demographic equity, and (5) advanced AI paradigms including transformers, LLMs, and reinforcement learning. Table II provides a consolidated comparison of all twelve articles.

B. Porto (2024) — ML & NLP for ED Triage

Porto et al. [1] present a PRISMA 2020 systematic review analyzing how ML and NLP algorithms improve triage accuracy. Models reviewed included decision trees, SVMs, neural networks (LSTM, BERT), and NLP approaches. ML models consistently outperformed traditional triage scoring scales, but the review identified a complete absence of continuous wait time regression and limited explainability work.

C. Multiple Authors (2025) — Admission Prediction from Triage

This PRISMA-compliant SLR [2] evaluated ML models on ED triage data for predicting inpatient admissions. Seven studies with rigorous methodology achieved AUC-ROC of 0.81–0.93, with random forest and gradient boosting as top performers, confirming that triage features (vitals, ESI level, chief complaint) are strong predictors.

D. Wang et al. (2025) — XGBoost Fairness in ED Wait Times

Wang et al. [3] developed an interpretable XGBoost model for predicting prolonged ED wait times ($\geq$30 min) in ESI-3 patients. XGBoost achieved AUC $\approx$ 0.72, and SHAP identified triage ESI level and arrival hour as the top predictors. Performance disparities were found across race/ethnicity and insurance type, providing a direct methodological template and performance benchmark for our project.

E. Xie et al. (2022) — MIMIC-IV-ED Benchmark Suite

Xie et al. [4] created the first open-source benchmark dataset and pipeline for ED prediction using MIMIC-IV-ED ($\sim$400,000 ED visits). Gradient boosting and random forest outperformed all clinical scoring systems. This paper provides the preprocessing pipeline and benchmark structure our project builds upon.

F. Ricciardi et al. (2024) — ML for ED Length of Stay

Ricciardi et al. [5] evaluated ML classifiers for predicting prolonged ED LOS ($\geq$4 hours) on 496,172 admissions from Italy. Random Forest achieved best performance (AUC $\approx$ 0.80), validating that triage level, arrival time, and demographics are sufficient predictors—exactly our feature set.

G. Seo et al. (2024) — Wait Time Prediction with NLP

Seo et al. [6] developed ML models integrating structured triage data with NLP-processed chief complaint text on 271,143 ED patients. LightGBM achieved the best performance and a real-time clinical dashboard prototype was developed, directly modeling our target outcome (ED waiting time).

H. Wang, Sambamoorthi et al. (2025) — Interpretable ML for ED Wait Time

Wang et al. [7] compared five ML models for predicting prolonged ED wait times in ESI-3 patients using SHAP values and partial dependency plots. XGBoost and Random Forest achieved the best performance, directly confirming our chosen model family and interpretability approach. We extend this work with continuous regression, all 5 ESI levels, and fairness analysis.

I. Sulaiman et al. (2025) — ML + TE2Rules for ED LOS

Sulaiman et al. [8] applied a novel explainability framework combining ensemble ML with TE2Rules rule extraction on 10 minimal triage/demographic variables, advancing the explainability agenda beyond standard SHAP and motivating rule-based explanation as a future extension of our pipeline.

J. Wu et al. (2025) — ED LOS Continuous Regression

Wu et al. [9] developed ML regression models for predicting ED LOS as a continuous variable on 187,028 cases. LightGBM achieved the best performance with the lowest MAE and RMSE. Current emergency waiting time was identified as the single top predictor, directly supporting our core decision to treat ED LOS as a continuous regression problem.

TABLE II
SUMMARY COMPARISON OF ALL REVIEWED ARTICLES (N=12)

#	Paper (Short)	Year / Model(s)	Best Metric	Key Gap
1	ML & NLP for ED Triage (SLR) [1]	2024 / BERT, SVM, LSTM	AUC ∼0.85+	No wait time regression; no SHAP
2	Admission from Triage (SLR) [2]	2025 / RF, XGBoost, LR	AUC 0.81–0.93	Binary only; no public dataset
3	XGBoost Fairness in ED [3]	2025 / XGBoost + SHAP	AUC ≈0.72	Binary only; no ESI segmentation
4	MIMIC-IV-ED Benchmark [4]	2022 / GB, RF, MLP	Outperforms MEWS/NEWS	No wait time; no SHAP; no fairness
5	ML for ED Length of Stay [5]	2024 / RF, DT, SVM	AUC ≈0.80	Binary; no SHAP; no segmentation
6	Wait Time + NLP Prediction [6]	2024 / LightGBM + NLP	Best AUC-ROC overall	Single center; no SHAP; no fairness
7	Interpretable ML for ED [7]	2025 / XGBoost, RF, ANN, SVM	Best Acc/Recall/F1	Binary; ESI-3 only; no public data
8	ML + TE2Rules for LOS [8]	2025 / GB + TE2Rules	Best classification	Binary; single center; no regression
9	ED LOS Regression (China) [9]	2025 / LightGBM	Lowest MAE & RMSE	Single center; no SHAP; no fairness
10	AI Framework for Overcrowding [10]	2025 / TSiTPlus; XCM-Plus	MAE 4.19; R^2 0.44	Aggregate count, not patient-level
11	Intersectional Fairness in ED [11]	2025 / XGBoost (intersect.)	AUROC 0.85	Admission only; no regression
12	ED-Copilot: LLM + RL [12]	2024 / BioGPT + RL	Halved avg lab wait	Binary targets; no ESI segmentation

K. Vural et al. (2025) — AI Framework for ED Overcrowding

Vural et al. [10] evaluated 11 ML algorithms for predicting ED waiting room occupancy at 6-hour and 24-hour resolutions. TSiTPlus achieved MAE of 4.19 and R^2 of 0.56 for hourly prediction, extending ED prediction from individual patients to system-level capacity management.

L. Lett et al. (2025) — Intersectional Fairness in ED Admissions

Lett et al. [11] compared intersectional versus marginal fairness debiasing strategies in ED admission prediction using MIMIC-IV-ED. AUROC was 0.85 for both baseline and intersectionally debiased models, with intersectional debiasing reducing subgroup calibration error by 22.3% versus 11.3% for marginal debiasing, directly motivating the fairness analysis dimension of our project.

M. Sun et al. (2024) — ED-Copilot: LLM + RL

Sun et al. [12] introduced ED-Copilot, combining BioGPT with a reinforcement learning agent to recommend lab tests that minimize ED wait time. The system reduced average laboratory wait time by approximately 50% (∼4 hours to ∼2 hours), representing the most advanced AI paradigm in the review.

IV. SYNTHESIS AND IDENTIFIED RESEARCH GAPS

A. Overall Trends

A clear trend across all twelve articles confirms that gradient boosting ensemble methods (XGBoost, LightGBM, Random Forest) consistently outperform both traditional clinical scoring systems and simpler linear models. Growing emphasis on SHAP-based interpretability, consolidation around public benchmark datasets, and increasing attention to algorithmic fairness are the three other major trends observed.

B. Identified Research Gaps

The following gaps recur systematically:

- **Continuous regression:** Ten of twelve studies use binary classification. Only Wu et al. [9] and Seo et al. [6] model wait time as a continuous variable.
- **Triage-level segmentation:** No study reports performance broken down by all five ESI triage levels.
- **SHAP interpretability:** Only four of twelve studies apply SHAP; eight use no interpretability tool.
- **Public dataset reproducibility:** Nine of twelve studies use proprietary single-center data.
- **Fairness analysis:** Only two of twelve studies formally assess algorithmic fairness.

V. RESEARCH QUESTIONS

Based on our literature review, the following research questions guide our project:

1) **RQ1 (Prediction Accuracy):** Can regression-based gradient boosting models predict ED LOS as a continuous variable (in minutes) with lower error than binary classification baselines?
2) **RQ2 (Triage Segmentation):** Do model performance and key predictive features vary significantly across ESI triage severity levels (1–5)?
3) **RQ3 (Interpretability):** Which features most strongly drive ED LOS predictions according to SHAP value analysis, and do these align with the literature?
4) **RQ4 (Fairness):** Does predictive performance differ significantly across demographic subgroups (race/ethnicity, gender, age group)?

VI. Data Collection and Wrangling

A. Dataset Description

We use the Hospital Emergency Room Visits dataset (Kaggle, CC0 license), structured to mirror the MIMIC-IV-ED benchmark [4]. The raw dataset contains **9,216 ED visits** and **17 features**, including structured triage data, vital signs, patient demographics, arrival context, and disposition outcomes. Table III presents the full data dictionary.

TABLE III
Data Dictionary (Raw Dataset, N = 9,216 rows, 17 features)

#	Feature	Type	Description
1	`patient_id`	int64	Unique visit identifier
2	`age`	int64	Patient age (years)
3	`gender`	object	Biological sex (M / F)
4	`race`	object	Self-reported race/ethnicity
5	`arrival_hour`	int64	Hour of arrival (0–23)
6	`arrival_transport`	object	Mode of arrival
7	`acuity`	int64	ESI triage level (1–5)
8	`pain_score`	float64	Pain score (0–10)
9	`heartrate`	float64	Heart rate (bpm)
10	`sbp`	float64	Systolic BP (mmHg)
11	`dbp`	float64	Diastolic BP (mmHg)
12	`temperature`	float64	Body temperature (°F)
13	`o2sat`	float64	O_2 saturation (%)
14	`resprate`	float64	Respiratory rate (br/min)
15	`disposition`	object	Discharge disposition
16	`was_admitted`	int64	1 if admitted, 0 otherwise
17	**`ed_los_minutes`**	float64	**TARGET: ED LOS (min)**

B. Target Variable

The target variable is `ed_los_minutes`, a continuous measure of total time spent in the ED in minutes. Table IV summarizes its distribution. The distribution is strongly right-skewed (mean 189.8 min, median 153.7 min), with extreme values up to 2,644.9 minutes representing complex or boarding cases. Log-transformation (`log1p`) normalizes the distribution, justifying its use as a preprocessing step for Milestone 3 regression modeling.

TABLE IV
Target Variable: `ed_los_minutes` (N = 9,216)

Statistic	Value
Mean	189.8 min
Std	141.7 min
Min	8.6 min
25th percentile	95.6 min
Median (50th)	153.7 min
75th percentile	241.6 min
Max	2,644.9 min

C. Missing Value Analysis

Seven features contained missing values, all confined to vital signs—consistent with real-world ED triage settings. No feature exceeded 12% missingness so all were retained and imputed. Table V summarizes the missing value profile.

TABLE V
Missing Value Summary (Raw Dataset, Total = 5,156 cells, 3.3%)

Feature	Missing Count	Missing %
temperature	1,105	11.99%
resprate	921	9.99%
o2sat	829	9.00%
pain_score	737	8.00%
dbp	552	5.99%
sbp	552	5.99%
heartrate	460	4.99%

D. Data Cleaning and Preprocessing

The following wrangling steps were applied sequentially:

1) **Missing value imputation:** Zero rows were dropped due to missing target values (`ed_los_minutes` was complete). Numeric vital signs were imputed using median imputation. Categorical fields were imputed with their mode: gender → "F", race → "WHITE", arrival_transport → "WALK IN", disposition → "HOME". After imputation, remaining missing values = 0.
2) **Outlier detection and treatment:** IQR analysis identified 415 outliers (4.50%) in `ed_los_minutes` (upper fence: 460.8 min) and a combined 628 outliers across vital signs (0.44%–2.42%). Z-score analysis ($|Z| > 3$) confirmed 158 extreme LOS values (1.71%). Clinical reference bounds were applied: heart rate [30, 250] bpm; SBP [60, 280] mmHg; DBP [30, 180] mmHg; temperature [93, 107]°F; SpO_2 [50, 100]%; respiratory rate [4, 50] br/min; pain score [0, 10]; LOS [5, 4320] min. No values fell outside these clinical bounds after imputation.
3) **Label encoding:** Four categorical features were label-encoded: gender ($F = 0$, $M = 1$); race (ASIAN=0, BLACK/AFRICAN AMERICAN=1, HISPANIC=2, OTHER=3, WHITE=4); arrival_transport (AMBULANCE=0, OTHER=1, REFERRED=2, WALK IN=3); disposition (ADMITTED=0, HOME=1, LEFT WITHOUT BEING SEEN=2, OBSERVATION=3, TRANSFER=4).
4) **Feature engineering:** Five new features were derived: *shock_index* (HR/SBP; mean=0.66, std=0.19, range: 0.21–1.97), *pulse_pressure* (SBP−DBP; mean=50.1 mmHg), *arrival_shift* (Day: 07:00–14:59; Evening: 15:00–22:59; Night: 23:00–06:59), *is_weekend* (25.0% of visits), and *age_group* (Pediatric ≤18; Young Adult 19–40; Middle Age 41–65; Elderly >65).
5) **Normalization:** StandardScaler was applied to all 19 numeric model features (mean ≈ 0, std ≈ 1), consistent with Wu et al. [9] and Wang et al. [3].

E. Before vs. After Wrangling Summary

Table VI compares the dataset before and after the full wrangling pipeline. The cleaned dataset

(`team3_ed_waittime_cleaned.csv`) contains **9,216 records** and **26 features** with zero remaining missing values.

TABLE VI
DATASET BEFORE VS. AFTER WRANGLING

Metric	Before	After
Rows	9,216	9,216
Columns	17	26
Missing cells	5,156	0
Missing %	3.3%	0.0%
Numeric features	13	22
Categorical features	4	5
Target mean (min)	189.8	189.8
Target std (min)	141.7	141.7
Target min (min)	8.6	8.6
Target max (min)	2,644.9	2,644.9

VII. EXPLORATORY DATA ANALYSIS

A. Target Variable Distribution

The raw `ed_los_minutes` distribution is strongly right-skewed: most visits fall under 600 minutes (mean 189.8 min, median 153.7 min) with a long tail extending to 2,644.9 minutes representing complex or boarding cases. Log-transformation (`log1p`) brings the distribution close to normality, suggesting this transformation will benefit regression modeling in Milestone 3. The 25th–75th percentile range is 95.6–241.6 minutes.

B. Vital Sign Distributions

Table VII presents descriptive statistics for key triage vital signs after imputation. Heart rate is approximately normally distributed (mean 82.1 bpm). Systolic BP centers near 128.2 mmHg. Oxygen saturation is left-skewed—most patients present with normal SpO_2 (95–100%) but a clinically important minority show dangerous desaturation (minimum 88.2%). Respiratory rate is near-normally distributed at 17.0 br/min. Shock index (mean 0.66) confirms that most patients are hemodynamically stable at triage, with a visible proportion exceeding the SI > 1.0 shock threshold.

TABLE VII
VITAL SIGN DESCRIPTIVE STATISTICS (POST-WRANGLING, N = 9,216)

Feature	Mean	Std	Min	Median	Max
heartrate	82.1	17.6	40.0	82.1	149.4
sbp	128.2	21.1	70.0	128.4	205.8
dbp	78.0	13.6	40.0	78.0	130.0
temperature	98.4	1.0	95.0	98.4	102.5
o2sat	97.3	2.1	88.2	97.5	100.0
resprate	17.0	3.7	8.0	17.0	33.8
pain_score	4.8	2.9	0.0	5.0	10.0
shock_index	0.66	0.19	0.21	0.64	1.97

C. Feature Correlation Analysis

Pearson correlation analysis reveals that *acuity* has the strongest association with `ed_los_minutes` ($r = -0.38$), confirming that lower ESI level numbers (higher severity) correspond to longer LOS. All other features show very weak linear correlations ($|r| \leq 0.02$), including temperature ($r = -0.02$), shock index ($r = 0.01$), and heart rate ($r = 0.01$). This confirms that ED LOS is a multivariate nonlinear problem where ensemble models capturing feature interactions are essential. The strong correlation between SBP and DBP motivates the engineered *pulse_pressure* feature to reduce multicollinearity.

D. ED LOS by ESI Triage Level

Table VIII shows that LOS varies substantially across ESI levels, confirming that a single pooled model would obscure important subgroup dynamics. ESI Level 1 patients have the longest mean LOS (315.7 min) due to intensive workups, while ESI Level 5 patients have the shortest (91.4 min). ESI Level 3 accounts for the largest proportion of visits (4,184 of 9,216; 45.4%), consistent with Wang et al. [3]. This directly supports our Milestone 3 plan for triage-stratified modeling.

TABLE VIII
ED LOS BY ESI TRIAGE LEVEL

ESI Level	N	Mean LOS (min)	Median LOS (min)
1 (Most Acute)	179	315.7	252.7
2	1,159	265.2	225.0
3	4,184	215.9	179.6
4	2,706	145.0	121.5
5 (Least Acute)	988	91.4	75.9
Overall	**9,216**	**189.8**	**153.7**

E. Temporal Patterns

Visit volume peaks during daytime hours (10:00–20:00). LOS is longer for early morning arrivals (02:00–06:00) as these patients tend to be more seriously ill. Evening arrivals coincide with the highest patient volumes, which may contribute to crowding. Table IX shows LOS by arrival shift. Night shift arrivals have the highest mean LOS (191.4 min) with a wider distribution, consistent with reduced staffing and a higher proportion of critically ill patients presenting after hours. These findings validate `arrival_hour` and `arrival_shift` as important predictors for Milestone 3, consistent with Wang et al. [7].

TABLE IX
ED LOS BY ARRIVAL SHIFT

Shift	N	Mean LOS (min)	Median LOS (min)
Day	3,074	192.0	155.1
Evening	3,103	186.2	151.2
Night	3,039	191.4	154.4

F. Arrival Mode and Disposition

Walk-in patients form the majority of arrivals (5,714; 62.0%), followed by ambulance (2,045; 22.2%), referred (933; 10.1%), and other (524; 5.7%). Ambulance arrivals show a higher mean LOS (192.5 min) than walk-in patients (189.2 min), consistent with their higher acuity. Walk-in patients' high volume with relatively shorter LOS reflects the proportion of lower-acuity visits. Home discharge dominates disposition (5,302; 57.5%), followed by admitted (2,632; 28.6%), observation (475; 5.2%), transfer (426; 4.6%), and left-without-being-seen (381; 4.1%).

G. Demographic Patterns and Fairness Signals

Table X shows mean LOS by race/ethnicity. White patients have the highest mean LOS (193.3 min) while Hispanic patients have the lowest (180.9 min), a difference of 12.4 minutes. By gender, males have a slightly higher mean LOS (190.9 min) than females (188.8 min), a difference of 2.1 minutes. Table XI shows LOS by age group: Young Adults have the longest mean LOS (193.5 min) while Elderly patients have the shortest (179.2 min). These preliminary disparities motivate the formal fairness analysis planned for Milestone 3, building on the methodology of Wang et al. [3] and Lett et al. [11].

TABLE X
MEAN ED LOS BY RACE/ETHNICITY

Race/Ethnicity	N	Mean LOS (min)
WHITE	5,061	193.3
BLACK/AFRICAN AMERICAN	1,874	188.0
HISPANIC	1,288	180.9
ASIAN	646	189.1
OTHER	347	183.8

TABLE XI
MEAN ED LOS BY AGE GROUP

Age Group	N	Mean LOS (min)
Pediatric ($\leq$18)	697	182.0
Young Adult (19–40)	3,113	193.5
Middle Age (41–65)	4,350	191.1
Elderly (>65)	1,056	179.2

VIII. MACHINE LEARNING PIPELINE AND INTERMEDIATE RESULTS

A. Feature Set

The final model feature set comprises 19 standardized variables: `age`, `heartrate`, `sbp`, `dbp`, `temperature`, `o2sat`, `resprate`, `pain_score`, `acuity`, `shock_index`, `pulse_pressure`, `arrival_hour`, `is_weekend`, `was_admitted`, `gender_encoded`, `race_encoded`, `arrival_transport_encoded`, `arrival_shift_encoded`, `age_group_encoded`.

B. Models Selected

Based on the literature's consistent findings, we evaluate:

- **Baseline:** Linear Regression, Ridge Regression
- **Tree-based:** Random Forest Regressor
- **Gradient Boosting:** XGBoost Regressor, LightGBM Regressor

C. Evaluation Metrics and Train/Test Split

Models are evaluated on MAE, RMSE, and R^2 on a held-out test set, consistent with Wu et al. [9]. An 80/20 chronological train/test split is applied to simulate real deployment conditions.

D. Expected Results and SHAP Analysis

Based on the correlation analysis ($r = -0.38$ for acuity) and literature findings, ESI acuity level is expected to be the dominant SHAP predictor, consistent with Wang et al. [3] and Wang et al. [7]. Arrival hour, shock index, and `was_admitted` are expected to rank among the top features. Gradient boosting models (XGBoost, LightGBM) are expected to outperform the linear baselines given the weak linear correlations and strong nonlinear interactions in the feature space. Full model results, SHAP summary plots, and patient-level force plots will be delivered in Milestone 3.

IX. NEXT STEPS TOWARD MILESTONE 3

1) **Model training and evaluation:** Train all five models on the 9,216-record cleaned dataset and report MAE, RMSE, and R^2.
2) **Hyperparameter tuning:** Apply grid search with 5-fold cross-validation to XGBoost and LightGBM.
3) **Triage-stratified modeling:** Train and evaluate separate models for each ESI level (1–5) and compare against the pooled model, addressing the gap absent from all twelve reviewed studies.
4) **Full SHAP analysis:** Generate population-level SHAP summary plots and patient-level force plots for clinical interpretability.
5) **Fairness analysis:** Compute MAE and RMSE broken down by race/ethnicity (5 groups), gender (2 groups), and age group (4 groups) to quantify the disparities identified in Section VII-F.
6) **Final paper:** Complete the research paper with full results, discussion, and conclusion sections.

X. CONCLUSION

The twelve reviewed articles confirm that gradient boosting ensemble methods are well-suited to predicting ED wait times, and no study to date has combined continuous wait time regression, ESI-level segmentation, SHAP interpretability, demographic fairness analysis, and a publicly accessible dataset in a single coherent framework. Our wrangling pipeline successfully processed 9,216 ED visits from 17 raw features to 26 engineered features with zero remaining missing values. EDA confirms that ESI acuity level ($r = -0.38$ with LOS) is the dominant predictor, with mean LOS ranging from 315.7

min (ESI 1) to 91.4 min (ESI 5). Preliminary demographic disparities—up to 12.4 minutes across race/ethnicity groups—motivate the formal fairness analysis in Milestone 3. Our project builds directly on the methodological foundations of Xie et al. [4], Wang et al. [3], Wu et al. [9], Lett et al. [11], and Wang et al. [7] to deliver a reproducible, interpretable, and fair ED wait time prediction system.

ACKNOWLEDGMENTS

This work was conducted as part of the CSCI-7090 Data Science and Machine Learning course project, Team 3, Spring 2026, Georgia Southern University.

REFERENCES

[1] B. M. Porto, "Improving triage performance in emergency departments using machine learning and natural language processing: a systematic review," *BMC Emerg. Med.*, vol. 24, no. 1, p. 219, 2024. [Online]. Available: https://doi.org/10.1186/s12873-024-01135-2

[2] Multiple authors, "Predicting inpatient admissions from emergency department triage using machine learning: a systematic review," *JACEP Open*, Elsevier, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2949761225000045

[3] Wang et al., "Evaluating fairness of machine learning prediction of prolonged wait times in emergency department with interpretable eXtreme gradient boosting," *PLOS Digit. Health*, vol. 4, no. 3, 2025. [Online]. Available: https://doi.org/10.1371/journal.pdig.0000751

[4] F. Xie, J. Zhou, J. W. Lee et al., "Benchmarking emergency department prediction models with machine learning and public electronic health records," *Sci. Data*, vol. 9, p. 658, Oct. 2022. [Online]. Available: https://doi.org/10.1038/s41597-022-01782-9

[5] C. Ricciardi et al., "Evaluation of different machine learning algorithms for predicting the length of stay in the emergency departments: a single-centre study," *Front. Digit. Health*, vol. 5, p. 1323849, Jan. 2024. [Online]. Available: https://doi.org/10.3389/fdgth.2023.1323849

[6] H. Seo et al., "Prediction of hospitalization and waiting time within 24 hours of emergency department patients with unstructured text data," *Health Care Manag. Sci.*, vol. 27, pp. 114–129, 2024. [Online]. Available: https://doi.org/10.1007/s10729-023-09660-5

[7] H. Wang, N. Sambamoorthi, D. Sandlin, and U. Sambamoorthi, "Interpretable machine learning models for prolonged emergency department wait time prediction," *BMC Health Serv. Res.*, vol. 25, p. 403, Mar. 2025. [Online]. Available: https://doi.org/10.1186/s12913-025-12535-w

[8] W. A. Sulaiman et al., "Leveraging machine learning and rule extraction for enhanced transparency in emergency department length of stay prediction," *Front. Digit. Health*, vol. 6, p. 1498939, Feb. 2025. [Online]. Available: https://doi.org/10.3389/fdgth.2024.1498939

[9] W. Wu et al., "Development of an emergency department length-of-stay prediction model based on machine learning," *World J. Emerg. Med.*, vol. 16, no. 3, pp. 220–224, 2025. [Online]. Available: https://doi.org/10.5847/wjem.j.1920-8642.2025.048

[10] O. Vural et al., "An artificial intelligence-based framework for predicting emergency department overcrowding: development and evaluation study," *JMIR Med. Inform.*, vol. 13, p. e73960, 2025. [Online]. Available: https://doi.org/10.2196/73960

[11] E. Lett et al., "Intersectional and marginal debiasing in prediction models for emergency admissions," *JAMA Netw. Open*, vol. 8, no. 5, p. e2512947, May 2025. [Online]. Available: https://doi.org/10.1001/jamanetworkopen.2025.12947

[12] L. Sun et al., "ED-Copilot: reduce emergency department wait time with language model diagnostic assistance," in *Proc. 41st Int. Conf. Mach. Learn. (ICML)*, PMLR, vol. 235, pp. 46942–46956, Jul. 2024. [Online]. Available: https://proceedings.mlr.press/v235/sun24a.html

[13] A. Johnson et al., "MIMIC-IV-ED (Version 2.2)," PhysioNet, Jan. 2023. [Online]. Available: https://doi.org/10.13026/5ntk-km72

[14] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.